Lecture notes for Applies statistics, Probability theory

Lecture 3.

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Presentation Overview

Discrete random variables (DRV)

Random variable: a quantity depending on randomness. Discrete random variable: the range is countable.

Example

Toss a fair coin, if it shows H, then we get 1$, if it shows T, then we pay 1$. Then

$$P(X = 1) = \frac{1}{2}, P(X = -1) = \frac{1}{2}$$

is its distribution.

Definition

Given a sample space Ω, a set of events $\mathscr{F}$, a probability function P, and a countable set of real numbers D, a discrete random variable is a function with domain Ω and range D.

The distribution of a discrete random variable

Definition

Let $x_1, x:2, \cdots$ be the range of X. Denote by A_i the event $\{X = x_i\}, i = 1, 2, \ldots$. Then $A_i, i = 1, 2, \ldots$, is a partition of the sample space. So

$$p_i = P(A_i) = P\{X = x_i\}, i = 1, 2, \ldots,$$

is a discrete distribution, i.e. $p_i \geq 0, i = 1, 2, \ldots$, and

$$\sum_{i=0}^{\infty} p_i = 1$$

The sequence $p_1, p_2, \ldots$ is called the distribution of X

Hypergeometric distribution

In a box there are M red and $N-M$ white balls $(M<N)$. Choose n balls **without replacement** $(n<N)$. Denote by X the number of red balls chosen. Then X has hypergeometric distribution:

$$h_k = P(X=k) = \frac{\binom{M}{K}\binom{N-M}{n-k}}{\binom{N}{n}},$$

where $k = \max\{n-N+M, 0\}, \ldots, \min\{n, M\}$.

Binomial distribution

In a box there are M red and $N-M$ white balls $(M<N)$. Choose n balls with **replacement**! Denote by X the number of red balls chosen. Then X has binomial distribution:

$$b_k = P(X=k) = \binom{n}{k}\left(\frac{M}{N}\right)^k \left(1-\frac{M}{N}\right)^{n-k}, k=0,1,\ldots,n.$$

General form of the binomial distribution

Repeat an experiment n times. Consider an event A in the experiment, $P(A)=p$. Denote X the number of occurrences of A. Then the distribution of X is

$$b_k = P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, k=0,1,\ldots,n.$$

Poisson distribution

We say that X has Poisson distribution with parameter λ if

$$p_k = P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, k = 0, 1, 2, \ldots,$$

where $\lambda > 0$ is a constant.

Example

Expected value of DRVs

Example

Roll a fair die. If it shows k, then we win k \$. The distribution of our random variable is

$$P(X=1)=\frac{1}{6},\ldots,P(X=6)=\frac{1}{6}.$$

The average of our win is $\mathbb{E}(X)=(1+2+3+4+5+6)/6=3.5$. However, if we play with a manipulated die, then the average win will be another value. If

$$P(X=1)=\frac{1}{12},P(X=2)=\frac{1}{6},P(X=3)=\frac{1}{6},$$

$$P(X=4)=\frac{1}{6},P(X=5)=\frac{1}{6},P(X=6)=\frac{1}{4},$$

then the average will be $\mathbb{E}(X)=\frac{1}{12}\cdot 1+\frac{1}{6}\cdot 2+\frac{1}{6}\cdot 3+\frac{1}{6}\cdot 4+\frac{1}{6}\cdot 5+\frac{1}{4}\cdot 6\approx 3.91$

Expected value of DRVs

Definition

Let $p_k = P(X = x_k), k = 1,2,\ldots,$ be the distribution of X. Assume that the series $\sum_k p_k x_k$ is absolutely convergent. Then the number

$$\mathbb{E}X = \sum_{k=1}^{\infty} p_k x_k$$

is called the expectation of X .

Special expected values

- **Binomial distribution:** $\mathbb{E} = n \cdot p$
- **Hypergeometric distribution:** $\mathbb{E} = \frac{Mn}{N}$
- **Poisson distribution:** $\mathbb{E} = \lambda$

Expected value of DRVs

Theorem

The expectation is linear, that is if $\mathbb{E}X$ and $\mathbb{E}Y$ exist and are finite, c is a constant, then

- $\mathbb{E}(X+Y)$ exists and it is finite and $\mathbb{E}(X+Y)=\mathbb{E}X+\mathbb{E}Y$;
- $\mathbb{E}(cX)$ exists and it is finite and $\mathbb{E}(cX)=c\mathbb{E}X$

Theorem

Let the distribution of X be $p_k = P(\xi = k), k = 1, 2, \ldots$. Let $f : \mathbb{R} \to \mathbb{R}$ be a function and $Y = f(\xi)$. Then

$$\mathbb{E}Y = \sum_{k=1}^{\infty} p_k f(x_k),$$

if either $\sum_{k=1}^{\infty} p_k f(x_k)$ is absolutely convergent of $\mathbb{E}Y$ is finite.

Example

$$\mathbb{E}X^2 = \sum_{k=1}^{\infty} p_k x_k^2$$

Variance of DRVs

The variance is the expectation of the squared deviation of a random variable from its mean. Informally, it measures how far a set of numbers is spread out from their average value.

Definition

Let X be a random variable, assume that $EX = m$ exists and it is finite. Then

$$\mathrm{Var}X = \mathbb{E}(X - m)^2$$

is called the variance of X .

Variance of DRVs

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Theorem

Let $\mathrm{Var}X < \infty$, then

$$\mathrm{Var}X = \mathbb{E}X^2 - \mathbb{E}^2X,$$

where E^2X is the abbreviation of $(EX)^2$.

Variance of DRVs

Properties of the variance

Theorem

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var} X$$

for all numbers $a, b \in \mathbb{R}$.

Steiner's formula

For any number a

$$\mathrm{Var} X = \mathbb{E}(X - a)^2 - (\mathbb{E}X - a)^2,$$

$$\mathbb{E}(X - a)^2 \geq \mathrm{Var} X.$$

In the above inequality we have equality if and only if $a = \mathbb{E}X$

Cumulative Distribution function (CDF)

Let X be a discrete random variable with distribution $P(X = x_i) = p_i, i = 1, 2, \ldots$. Then its CDF is $F_X(x) = P(X < x) = \sum_{\{i: x_i < x\}} P(X = xi) = \sum_{\{i: x_i < x\}} p_i$. Therefore in this particular case the CDF is a step function 'jumping' p_i at the point x_i.

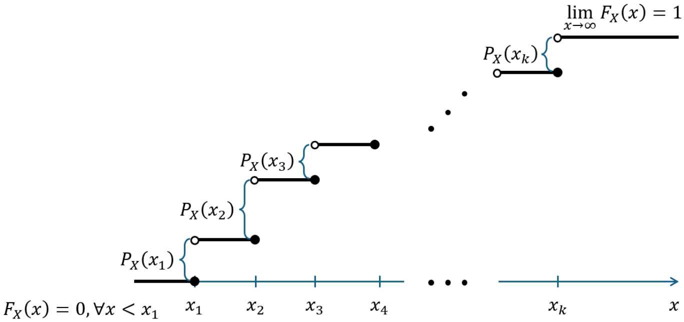

Fig. 1: CDF of a discrete random variable

Properties of CDF

Theorem

The function $F : \mathbb{R} \to \mathbb{R}$ is a CDF if and only if

I. F is monotone increasing,

II. F is left continuous,

III. $\lim\limits_{x\to\infty} F(x) = 1$, $\lim\limits_{x\to-\infty} F(x) = 0$

Exercise

I rolled a die. Let X denote the result of this experiment. Find the CDF of X!

Exercise

I toss a coin twice. Let X be the number of observed heads. Find the CDF of X.

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Hint for the solution

Note the following:

1. $X \sim Binomial\left(2, \frac{1}{2}\right)$
2. The range of X is: $R_X = \{0, 1, 2\}$
3. $P(x=0) = \frac{1}{4}$, $P(x=1) = \frac{1}{2}$, $P(x=2) = \frac{1}{4}$
4. Remember: $F(t) = P(X < t)$
5. What if $x \leq 0$, $0 < x \leq 1$, $1 < x \leq 2$, $2 < x$.